From sample to fastq

Outline

- How library quality affects data recovery
- Sequencing costs
- Estimate cost for your own experiment

Requirements for library prep protocol for lcWGS

- To prepare libraries for hundreds of samples, we need a protocol that is
  - Cheap
  - Efficient
  - Reliable
- Sometimes robustness to sample degradation is also important



Sequencing-Ready Fragment

Index 2 Primer



- P5 – complementary to Illumina flow cell oligo
- Indexing sequence 2
- Read 1 Sequencing Primer

Index 1 Primer



- Read 2 Sequencing Primer
- Indexing sequence 1
- P7 – complementary to Illumina flow cell oligo

Image adapted from the Nextera XT DNA Library Prep Kit Reference Guide (© 2017 Illumina, Inc.)

Unique vs. combinatorial dual index barcodes



Sequencing-Ready Fragment

Index 2 Primer



- P5 – complementary to Illumina flow cell oligo
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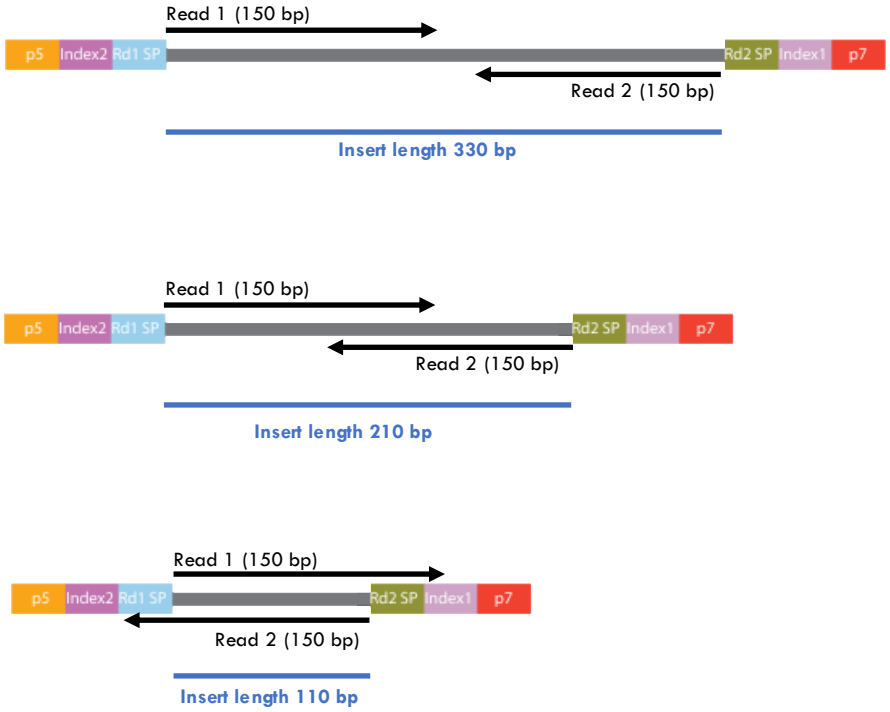
Index 1 Primer



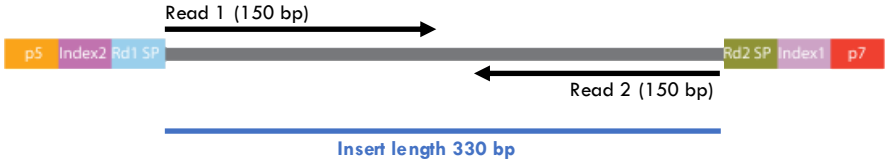
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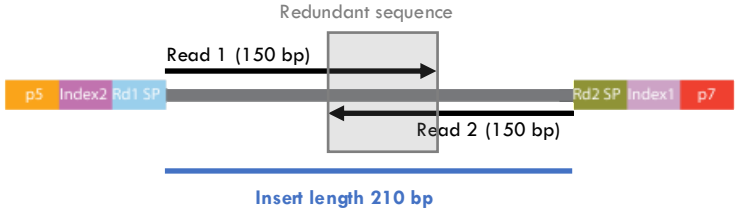
Image adapted from the Nextera XT DNA Library Prep Kit Reference Guide (© 2017 Illumina, Inc.)

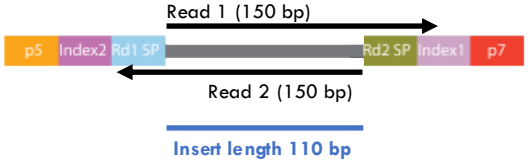
Insert length relative to read length



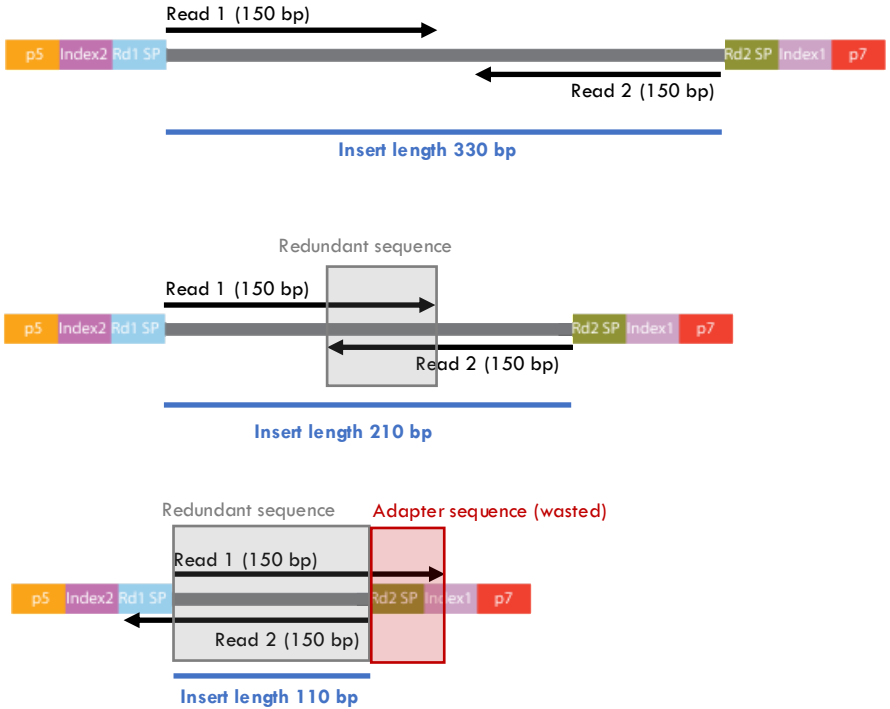
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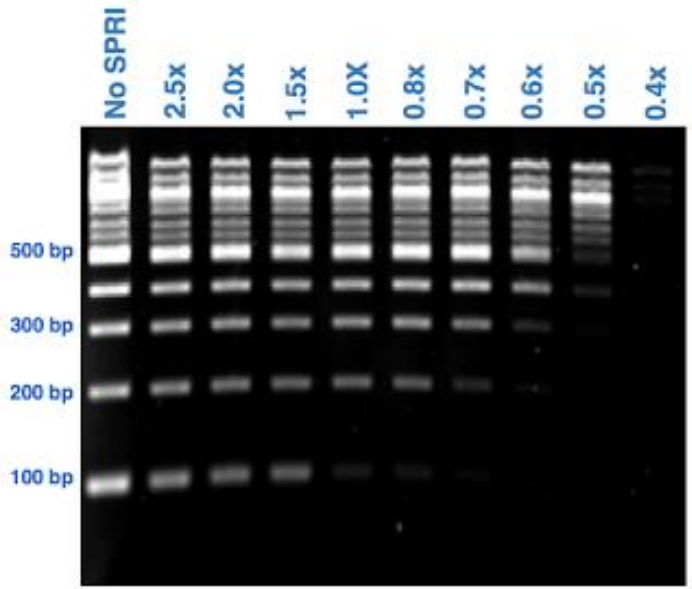


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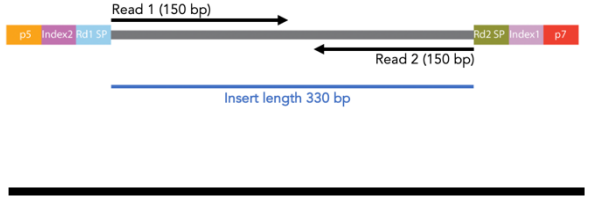
Size selection with Ampure beads

Tune the size distribution of your library fragments to minimize “waste” of sequence due to paired-end overlap and adapter read-through



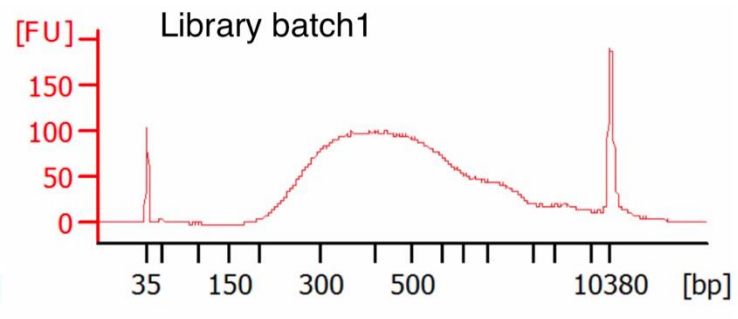
http://enseqlopedia.com/2012/04/how-do-spri-beads-work/

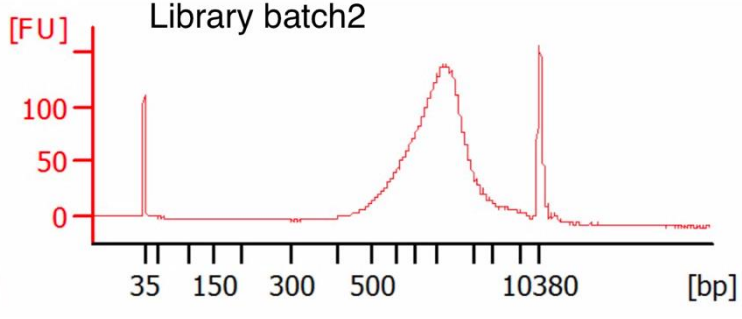
Ideally, we want all library fragments to be greater than the adapter length plus 2 x the read length (for PE)



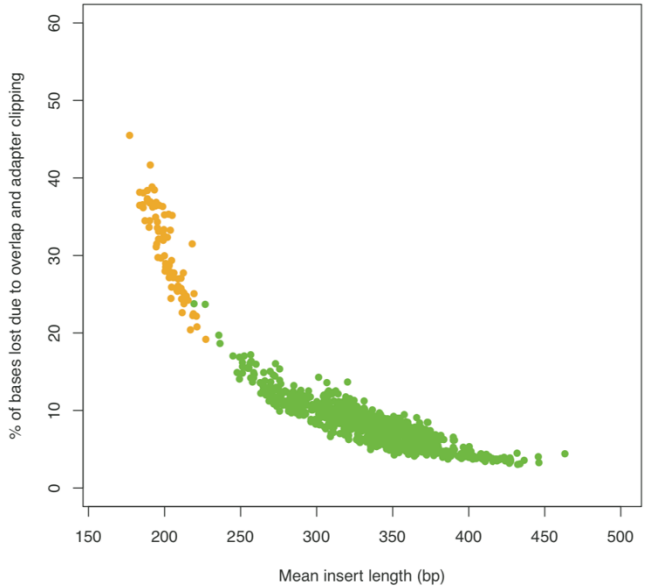
Ideal minimum fragment length

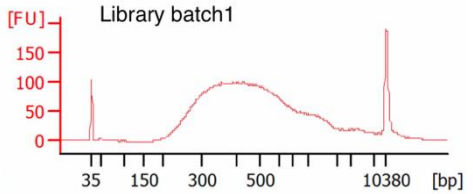
Two examples of our library pools

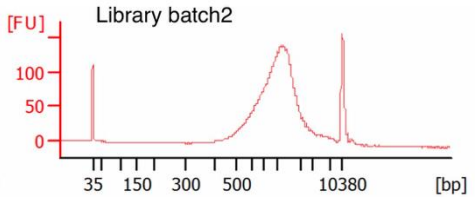




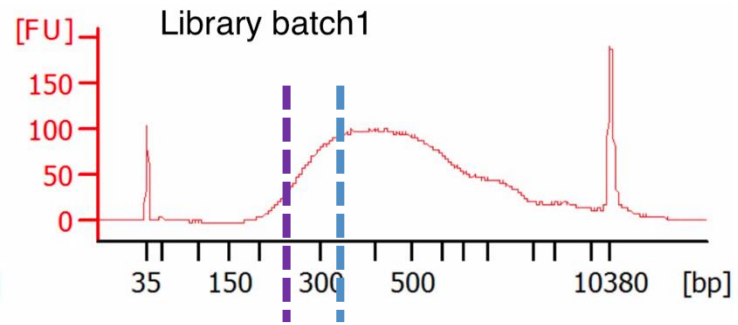
The library fragment size distribution can substantially influence the amount of data lost in data QC steps

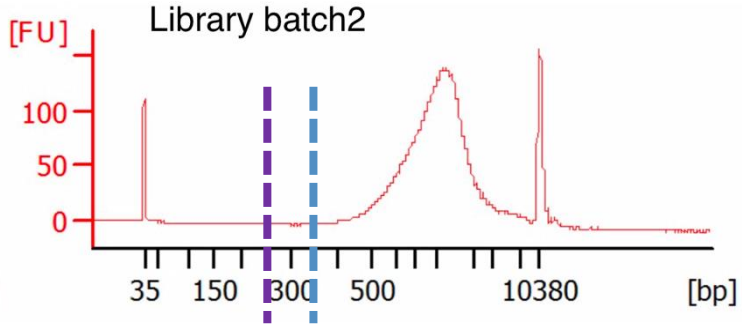






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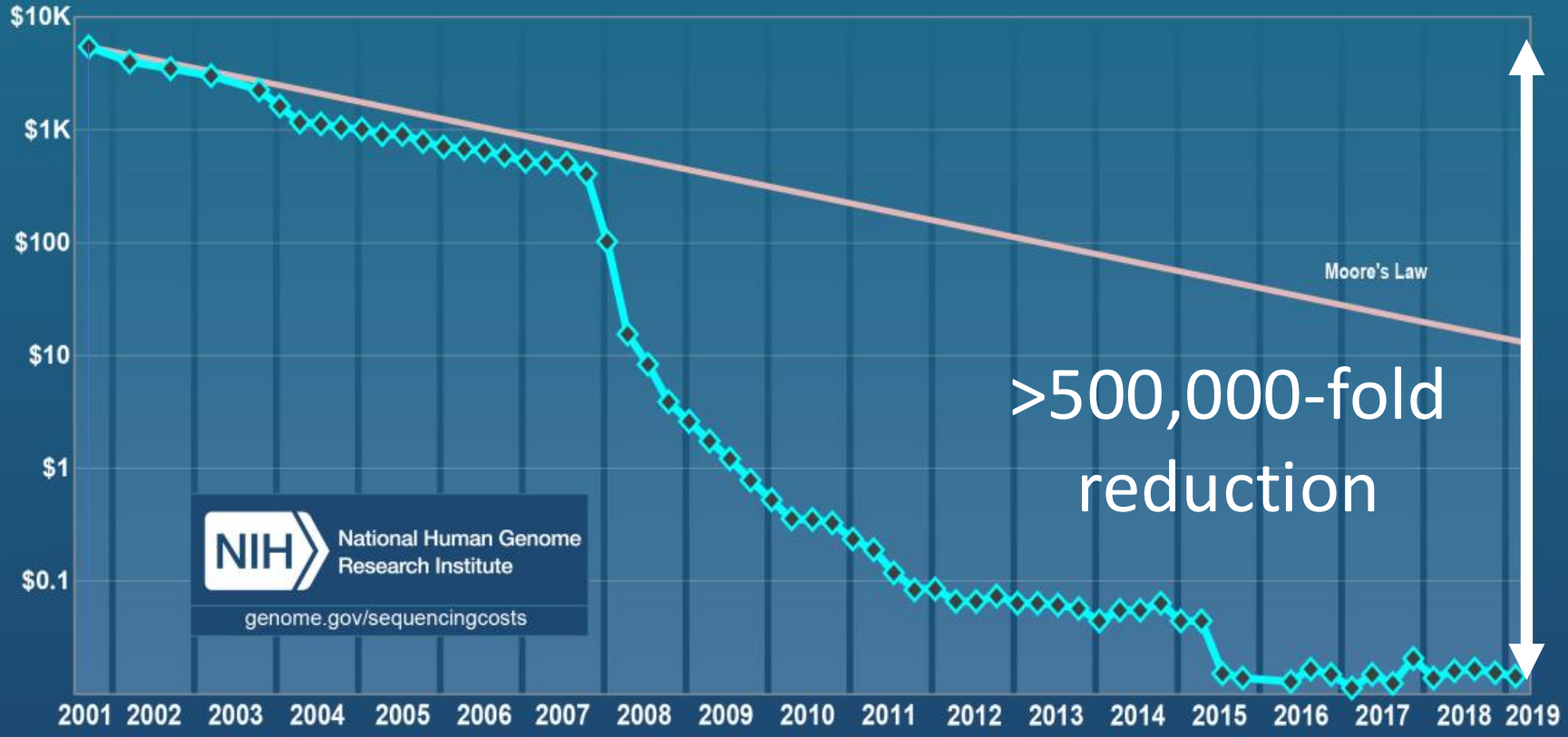




The length of Nextera adapters is 138 bp and libraries were sequenced with 2*125bp reads

- Minimum fragment length to avoid overlap 388bp
- Minimum fragment length to avoid adapter read-through 263bp

Cost per Raw Megabase of DNA Sequence



What is the current price for 2x sequencing of an Atlantic silverside (including library preparation)?

Genome size ~650 Mb

- $4
- $16
- $36
- $72
- $113

1 USD ≈ 1 EURO



https://www.menti.com/algjsqmxhf22

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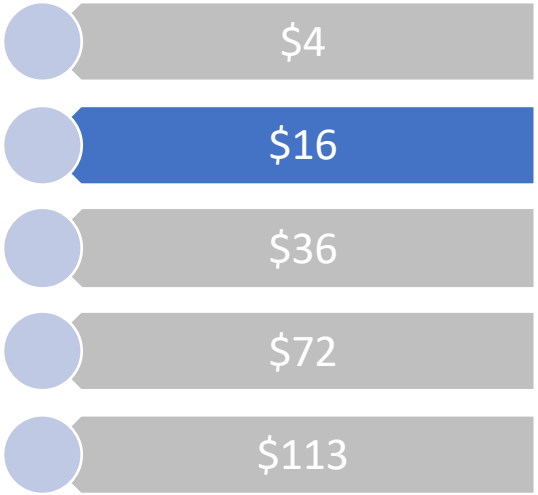
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Compare to:

$30 per sample for RADseq
$15 per sample for RADcapture

Meek and Larson. 2019. Mol Ecol Res



Important:
Sequencing cost scales with genome size

Exercise – how much will your experiment cost?

- Assumed costs:
  - Library preparation: $8 per sample
  - Sequencing: $4 per Gb (assuming a full NovaSeqX 25B lane)
  - Target coverage per sample: Expect to lose at least 30-50% of your data in filtering

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- Example: I would like to have 1x coverage for downstream analysis for 40 individuals from each of 5 populations (200 individuals total) of my favorite animal with a genome size of ~800 Mb

- Calculation: I will target 2x coverage raw sequencing. This means

  2 * 800 Mb/individual * 200 individuals = 320,000 Mb (320 Gb)

  My total cost is thus (320 Gb * $4/Gb) + (200 libraries * $8 per library) = **$2,880**